Course Project

The course project is a fundamental component of the Large-Scale Project Programming course and is designed to provide students with practical experience in analyzing, designing, implementing, and documenting a real-world software system.

Students are expected to apply the concepts learned throughout the course, including Software Development Life Cycle (SDLC), Object-Oriented Programming (OOP), Unified Modeling Language (UML), Database Design, and Software Documentation, to develop a professional software application.

Each group will select one project from the list of available projects. Every project represents a real-world software system and should be developed according to professional software engineering practices.

Project Objectives

By completing this project, students should be able to:

Analyze real-world software requirements.

Apply Software Engineering principles.

Design professional UML diagrams.

Design and implement a relational database.

Develop a complete software application using Object-Oriented Programming.

Work effectively as a software development team.

Prepare professional software documentation.

Present and demonstrate the developed system

Team Formation

Each project must be completed by a team consisting of 3–4 students.

Each team is allowed to choose one project only.

Project selection is based on a first-come, first-served basis.

Once a project has been assigned to a group, it cannot be changed

Project Instructions

The project must be developed by a team of 3–4 students.

All team members are expected to contribute equally.

The project must follow the Software Development Life Cycle (SDLC).

Object-Oriented Programming principles must be applied throughout the implementation.

A relational database must be used.

The application should provide a clear, user-friendly interface.

All submitted work must be original.

Copying projects from previous semesters, the Internet, GitHub, or other groups is strictly prohibited.

Every group member should be able to explain the part he/she developed.

Groups are encouraged to use GitHub to manage project versions and collaboration.

Submission Guidelines

Each group must submit the following:

Project Source Code

Project Report (PDF)

SQL Database File

Presentation Slides (PowerPoint or PDF)

Project Demonstration Video (Optional if requested)

GitHub Repository Link (if available)

All files should be compressed into one ZIP file and uploaded to the course platform before the submission deadline

Deadline for delivery 30-07-2026